

Silicon Core Harvesting is Visible in Supercomputer Telemetry: A Fleet-Scale Study of 1,960 POWER9 Sockets

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Abstract—Processor vendors sell partially defective dies by permanently disabling faulty units, a practice known as core harvesting. The resulting per-die disable maps are proprietary and are conventionally invisible to system owners. We show that the out-of-band (BMC/IPMI) sensor telemetry routinely collected by production supercomputers exposes these maps directly, and we recover them for all 1,960 processor sockets of CINECA’s Marconi100 from the public M100 ExaData release [1]. Every socket exposes exactly 16 of 24 possible per-core thermal sensors, and the eight absent sensors always form four complete *slice* pairs $(2k, 2k+1)$ —the POWER9 unit that shares an L2/L3 block [2], [3]—with zero exceptions. We observe 442 of the 495 combinatorially possible patterns, establishing that the map is per-die rather than a fixed SKU. Using a curveball null that preserves both per-socket and per-slice marginals, we show harvested slices cluster in sensor-index space ($z = -20.8$). Thermal analysis confirms that the two cores of a slice are physically abutted (residual correlation $r = +0.51$ within-slice versus $+0.29$ for cross-slice neighbours at identical index distance), but we find only weak evidence that the slice index itself is monotonic in physical die position ($p = 0.025$; a 3×4 grid fits the thermal coupling better than a linear arrangement). Finally, we detect a sharp procurement-lot boundary at rack 22 (Welch $t = 27.3$) separating two populations with markedly different harvesting statistics.

Index Terms—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance

I. INTRODUCTION

[TODO: write]. Framing: (i) core harvesting is standard industry practice but per-die maps are undocumented; (ii) HPC operators now collect holistic out-of-band telemetry [1], which we show carries an unintended side channel revealing manufacturing outcomes; (iii) contributions are a recovery method, the first fleet-scale characterisation of harvesting in a deployed Top-10 system, evidence of index-space clustering and of procurement-lot structure, and a thermal method for testing die-topology hypotheses.

II. BACKGROUND

A. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a 695 mm^2 die in 14 nm FinFET with 17 metal layers [3]. Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a slice, and each slice owns 512 kB of L2*

and 10 MB of L3 [3], [2]. IBM’s Hot Chips 28 presentation shows the 24 cores above $12 \times 10 \text{ MB}$ L3 blocks, attached to a 7 TB/s on-chip fabric at $256 \text{ GB/s} \times 12$ [2]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with 2×16 -core POWER9 at 2.6/3.1 GHz and $4 \times$ NVIDIA V100 [4], [5]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 1 shows the organisation annotated with the harvest rates we measure.

B. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9 TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20 s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, “Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23” [6]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. METHOD

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s, k] \in \{0, 1\}^{12}$ over slices $k = \lfloor \text{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \dots, 23\}$, so the index is a fixed physical-position identifier.

B. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [7], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. Row- and column-sum preservation is asserted on each of 200 draws.

C. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. RESULTS

A. Every socket is 16 cores; granularity is the slice

TABLE I
DISABLE-MAP STRUCTURE, 2020-04.

Nodes analysed	980
Sockets	1,960
Sockets reporting exactly 16 cores	1,960 (100%)
Sockets reporting 24 cores	0
Absent cores form complete $(2k, 2k+1)$ pairs	1,960 (100%)

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

B. The map is per-die, not per-SKU

442 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 7). The most common (slices 0–3) covers only 5.7% of sockets. Within a node the two sockets match in just 1.2% of cases, with mean overlap 1.648 slices against a shuffled null of 1.474 ± 0.026 ($z = +6.6$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.7% of sockets against a uniform expectation of 33.3% ($\chi^2 = 836.5$, $df = 11$, $p \ll 0.001$); Fig. 2 splits this by lot.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.243 versus 4.585 ± 0.016 ($z = -20.8$), and every significant co-harvest excess has $|\Delta k| \leq 2$ (Fig. 3): slices 0&1 ($z = +19.5$), 6&7 (+9.5), 8&9 (+8.7), 2&3 (+8.4), 4&5 (+7.9), 1&2 (+7.4), 10&11 (+7.1). All negative terms are distant pairs.

E. Slice siblings are physically abutted

Fig. 4 shows residual correlation against core index distance. At *identical* distance of 1, within-slice siblings reach $r = +0.51$ versus $+0.29$ for cross-slice neighbours; per-node the contrast between near (≤ 2) and far (≥ 12) pairs is $+0.45 \pm 0.20$ with 40/40 nodes positive (paired $t = +13.9$). The two cores of a slice are roughly twice as coupled as equally-numbered cores across a slice boundary, confirming both physical abutment and the slice as a physical unit.

F. The slice index is not established as the physical ordering

Free classical MDS on the pooled 24×24 matrix is dominated by noise (leading axis 20.5% of variance, correlation with core index only $+0.34$), so we instead score explicit candidate floorplans by correlating predicted physical distance against measured thermal coupling (Fig. 5). The identity 1×12 linear arrangement scores -0.271 , better than a random ordering ($z = -2.22$, $p = 0.025$) but *worse* than a 3×4 grid (-0.565); a hill-climbing search finds an ordering scoring -0.715 whose rank correlation with the identity ordering is only $+0.34$.

We therefore do not claim the sensor index is monotonic in physical die position. The index-space clustering of Fig. 3 is a robust empirical fact, but its interpretation as *spatial* defect clustering is not established by our data and must be stated as a hypothesis. **[TODO: obtain an annotated POWER9 die shot or OpenBMC sensor-ID mapping to settle this.]**

G. A procurement-lot boundary at node 440

Node IDs map to position as $\text{rack} = \lfloor \text{id}/20 \rfloor$ [6]. A changepoint scan maximises Welch t at rack 22 (node 440), $t = 27.3$ (Fig. 6).

TABLE II
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY.

Population	Sockets	$P(\text{slice } 0)$	Distinct patterns
Lot A (racks 0–21)	880	89.1%	224
Lot B (racks 22–48)	1,080	39.4%	414

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.568 vs 1.454 ± 0.008 , $z = +14.1$).

H. Temporal stability

Between 2020-04 and 2021-03, 96.7% of sockets have byte-identical active-core sets. All 65 changed sockets are consistent with hardware replacement: for node 115 socket 0, one set stops on 2021-03-03 and a different set appears on 2021-03-30, each containing exactly 16 cores.

V. RELATED WORK

[TODO: expand; the citations below are located but unverified.] Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [8], turbo-boost variation across nominally identical processors [9], and accelerator-level variability in GPU-rich systems [10]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [11] and spatial wafer-map models [12] describe clustered killer defects but are validated on fab data unavailable to system owners. To our knowledge no prior work recovers per-die harvesting maps from deployed-system telemetry, which is the gap this paper addresses.

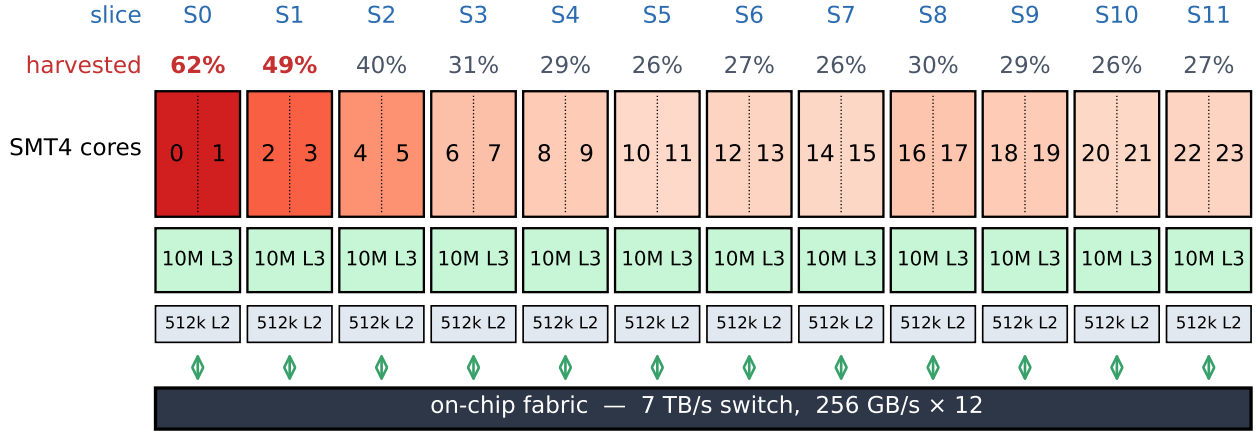


Fig. 1. POWER9 slice organisation (after [2]) annotated with the measured per-slice harvest rate across 1,960 Marconi100 sockets. Each slice holds two SMT4 cores sharing 512kB L2 and 10MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which we show is *not* established as the physical die ordering** (Sec. IV-F).

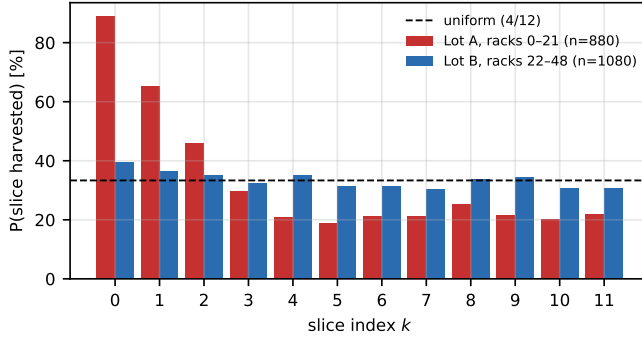


Fig. 2. Per-slice harvest probability by procurement lot.

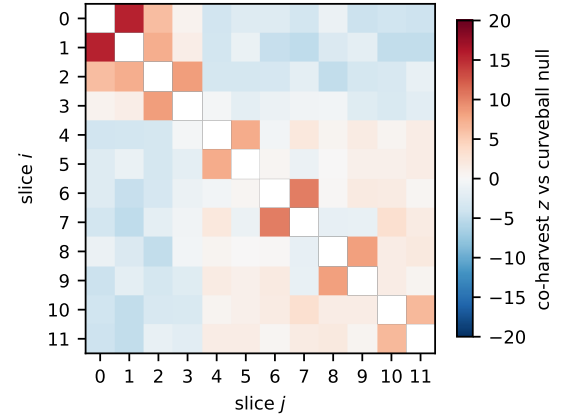


Fig. 3. Co-harvest z against the curveball null. Excess is confined to near-diagonal (index-adjacent) slice pairs.

VI. THREATS TO VALIDITY

- 1) **Sensor presence $\neq$ core existence.** Mitigated by 100% consistency across 1,960 sockets and two months, agreement with the published 16-core SKU [4], and the perfect slice-pair structure.
- 2) **Two months only.** **[TODO: extend to all 31 months for the full swap timeline.]**
- 3) **Physical mapping unresolved** (Sec. IV-F); the spatial interpretation of clustering is hypothesis, not result.
- 4) **Common-mode artefact**, addressed by same-distance contrasts.
- 5) **Single system**; findings may reflect CINECA procurement rather than POWER9 generally.

VII. CONCLUSION

[TODO: write.]

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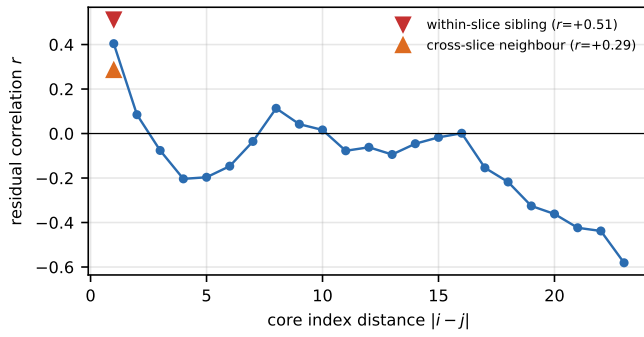


Fig. 4. Residual thermal correlation versus core index distance. The within-versus cross-slice contrast at distance 1 controls exactly for index gap.

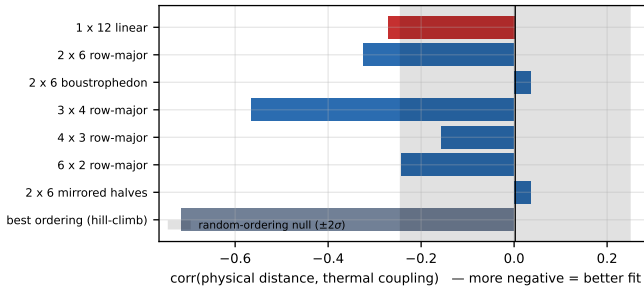


Fig. 5. Candidate floorplans scored against measured thermal coupling. The linear index ordering (red) beats a random ordering only marginally and is outperformed by a 3×4 grid.

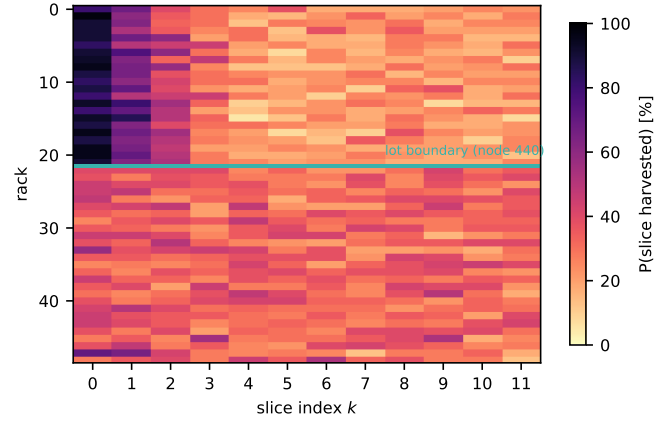


Fig. 6. Harvest probability by rack and slice. The lot boundary at rack 22 is visible as an abrupt change.

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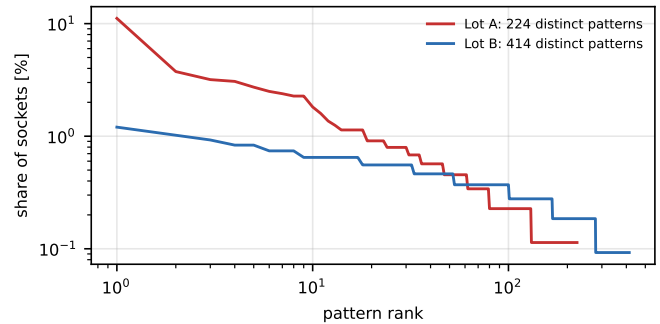


Fig. 7. Rank–frequency of disable patterns. Lot B is markedly more diverse.